

GI anatomy and physiology

Muluken N.

GI anatomy and physiology

- ❖ The gastrointestinal tract (GIT) is a muscular tube approximately 6 m in length with varying diameters.
- ❖ It stretches from the mouth to the anus.
- ❖ The GIT is highly specialized region of the body whose primary functions involve the processes of secretion, digestion and absorption.
- ❖ All of its functions are accomplished by complex neural, muscular, and hormonal processes.

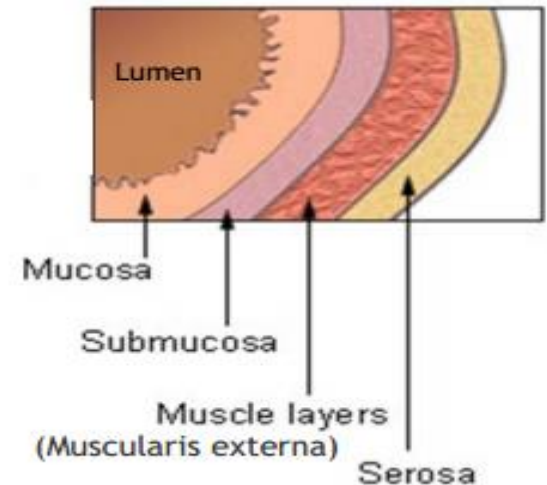
❖ The majority of the GI epithelium is covered by a layer of mucous.

- ✓ This is a viscoelastic translucent aqueous gel that is secreted throughout the GIT, acting as a protective layer and a mechanical barrier.

❖ The wall of the GIT is essentially similar in structure along its length, consisting of four principal histological layers

1. **Serosa:** an outer layer of epithelium

2. **Muscularis externa:** Contractions of these muscles provide the forces for movement of GI contents



3. **Submucosa:** containing some secretory tissue and supplied with blood vessels.

4. *Mucosa*: composed of three sublayers

A. Muscularis mucosa:

- thin sheet of smooth muscle 3-10 cells thick and separates the mucosa from the submucosa
- enhance contact between epithelium and contents of lumen
- increase surface area and rate of absorption

B. Lamina propria:

- contains capillaries, lymph vessels, nerve fibers and glands

C. Epithelium:

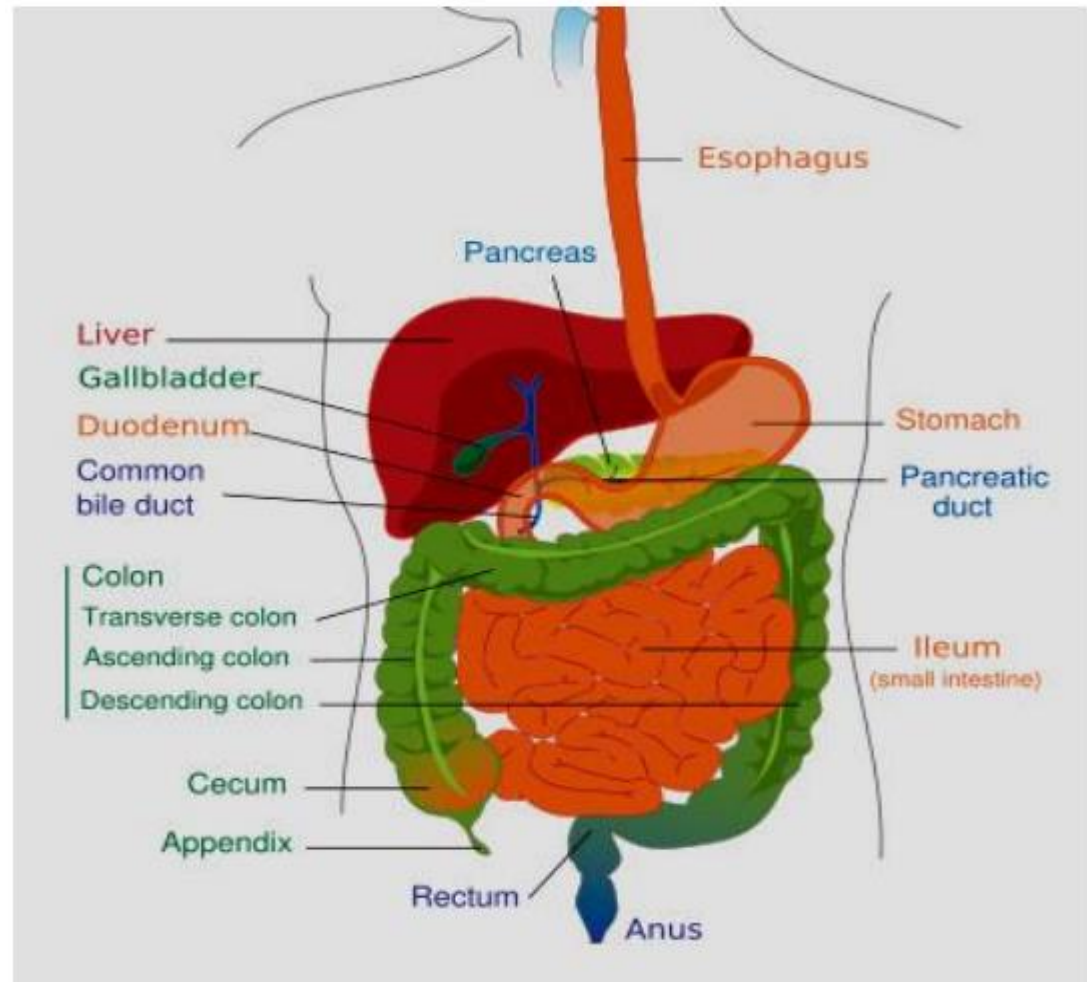
- a thin layer of tightly packed cells lining the GIT
- selective absorption, secretion, protective functions
- layer of mucus covers majority of GI epithelium

Esophagus

- links oral cavity with stomach
- composed of thick muscular layer ~ 25 cm long and 2 cm diameter
- Epithelial cell function is mainly protective:- simple mucous glands secrete mucus into the narrow lumen to lubricate food and protect the lower part of the esophagus from gastric acid.
- **pH of lumen ~ 5 - 6**
- Materials are moved down the esophagus by the act of swallowing.

- After swallowing, a single peristaltic wave of contraction passes down the length of the esophagus at the rate of 20-60 mm per second, speeding up as it progresses.
- In upright position, transit of materials is assisted by gravity

→ ***Esophageal
transit of DFs
is 10 - 14 s***



Stomach

- The most dilated part of the GI tract - two major functions:
 1. temporary reservoir for ingested food
 - Capacity: ~ 1.5 L in fed state
 - Fasting condition: not > 50 ml
 2. mix and reduce ingested components with the aid of gastric secretions into a thick fluid slurry

- Secretory cells form part of an extensive network of gastric glands which produce and secrete about **2L** of gastric fluid daily.

❑ These gastric secretions include:

- **acid** secreted by the parietal cells, which maintains the pH of the stomach b/n 1 and 3.5 in the fasted state;
- **the hormone gastrin**: its release is stimulated by peptides, amino acids and distension of the stomach;
- **pepsins**, which are secreted by the peptic cells in the form of its precursor pepsinogen;
 - Pepsins are peptidases which break down proteins to peptides at low pH.
- **mucus**: lines the gastric mucosa.
 - Mucus protects the gastric mucosa from autodigestion by the pepsin-acid combination.

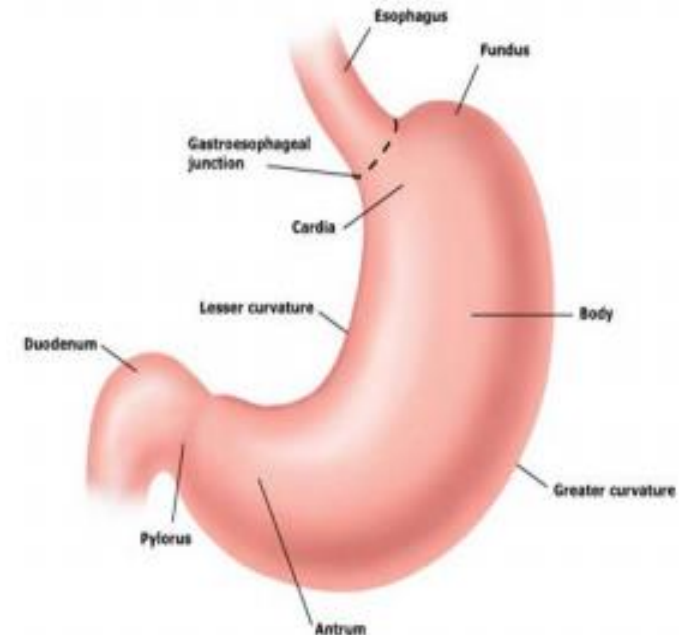
➤ Stomach is divided into four regions:

- ***Fundus, Body, Antrum, Pylorus***

➤ The proximal stomach, made up of the ***fundus*** and ***body*** serves as a reservoir for ingested material.

➤ The distal region, ***antrum*** and ***pylorus***, is the major site for mixing and pumping to accomplish gastric emptying.

- Mixing is intense and pressurized in the antral part of the stomach, a process of breaking down large food particles described as antral milling.



- The epithelial cells of the gastric mucosa represent one of the most rapidly proliferating epithelial tissues,
 - shed by the normal stomach at the rate of about half a million cells per minute.
- Very little drug absorption occurs
- High biopharmaceutical importance:
 - Gastric emptying can dictate drug absorption from SI .

Small intestine

- Longest (4 - 5 m) and most convoluted part
- 2 Main functions: - Digestion
- Absorption
- 3 parts: – Duodenum, Jejunum, Ileum:
- SI wall has rich network of ***blood*** and ***lymphatic*** vessels
- SI receives blood from ***superior mesenteric artery***
- Blood leaves SI via ***hepatic portal vein*** to liver and then to systemic circulation

- The surface epithelial cells of the SI are renewed rapidly and regularly.
- The luminal pH of the SI → **6 - 7.5**.
- The sources of the secretions that produce these pH values in the SI are:

A. *Brunner's glands*:

- located in the duodenum and secrete mucus-rich alkaline secretion
- provide alkaline condition for the activation of intestinal enzymes
- lubricate the intestinal walls

B. Intestinal cells: secrete mucus and enzymes

- hydrolases and proteases, continue the digestive process

C. Pancreatic secretions: the pancreas secretes about 1-2 L of pancreatic juice per day into the SI via a duct

- The components of pancreatic juice are sodium bicarbonate and enzymes.
- The enzymes consist of proteases, principally trypsin, chymotrypsin and carboxypeptidases. They are converted to their active forms in the lumen by the enzyme **enterokinase**.
- Lipase and amylase are both secreted in their active forms

D. *Bile*: secreted by hepatocytes in the liver and delivered to the duodenum.

- The main functions of bile are:
 - promoting efficient absorption of dietary fat by aiding its emulsification
 - provision of excretory pathways for degradation products.
- Bile is a complex aqueous mixture of organic solutes (bile acids, phospholipids...) and inorganic compounds.
- Bile acids are reabsorbed by an active process in the terminal ileum. They are returned to the liver via the hepatic portal vein and, as they have a high hepatic clearance, are re-secreted in the bile.
 - process is known as ***enterohepatic recirculation***.

→ SI has enormous surface area of ~ **200 m²** in adult

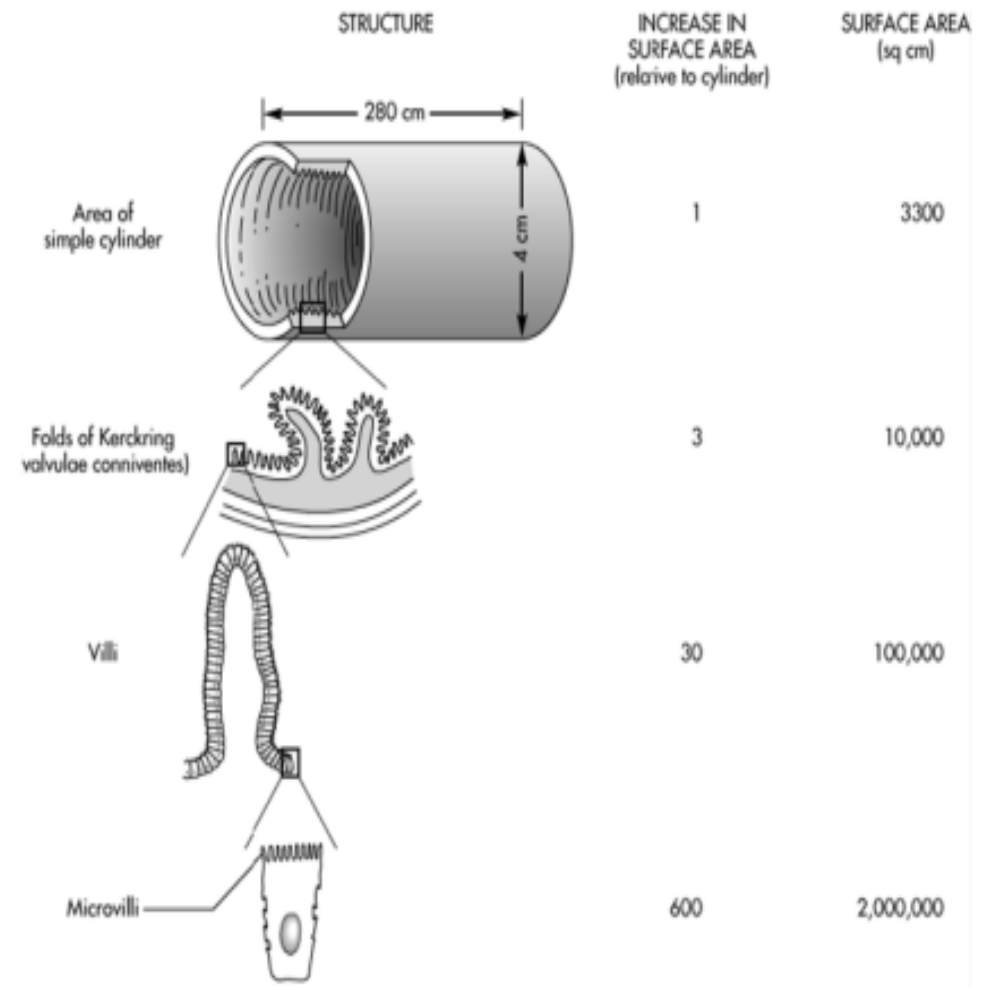
- ***Folds of Kerckring***: folds of SI mucous membrane that project within lumen
 - Well developed in duodenum and jejunum
- ***Villi***: finger-like projections into lumen
 - Estimated about 10-40 villi/ mm²
 - Well supplied with blood vessels
- ***Microvilli***: brush-like structures covering villi
 - there are ~ 600 microvilli protruding from each absorptive cell lining the villi.
 - largest increase in SA

– SI has significant biopharmaceutical importance:

→ Most nutrients and drugs absorbed from SI

➤ Relative to the surface area of a smooth cylinder, the folds, villi and microvilli

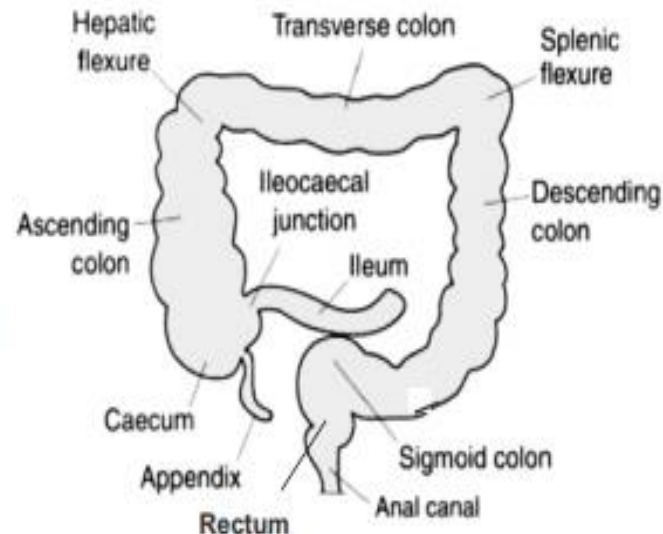
increase the effective SA by factors of 3, 30 and 600 respectively.



Large intestine

- Final part - ileocaecal junction to anus (~ 1.5m long)
- The large intestine has greater diameter (6cm)
- The wall of the ileum at the ileocaecal junction has a thickened muscular coat called the ***ileocaecal sphincter***

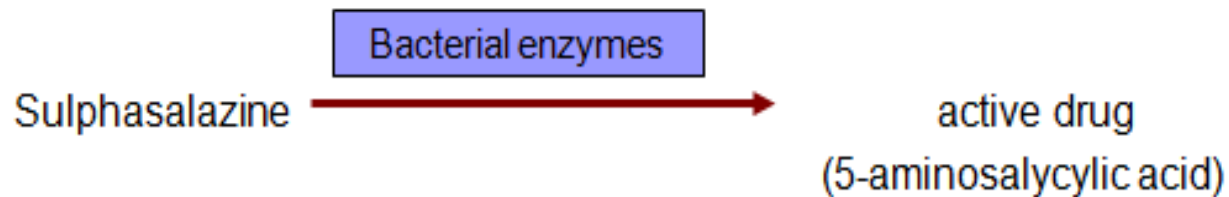
- caecum (~ 8.5 cm)
- ascending colon (~ 20 cm)
- transverse colon (> 45 cm)
- descending colon (~ 30 cm)
- sigmoid colon (~ 40 cm)
- rectum (~ 12 cm)



- From a functional point of view, the large intestine may be divided into two part.
 - The proximal half, concerned primarily with absorption
 - The distal half, concerned with storage and mass movement of fecal matter
 - storage and compaction of fecal material
- In humans, the large intestine usually receives about 500 mL of fluid-like food material (chyme) per day.

- Due to efficient water absorption, of the 500 mL normally reaching the large intestine, approx. 80 mL are eliminated from the gut as fecal material.
- Structurally, the LI is similar to the SI, but the luminal surface epithelium of the former lacks specialized villi.
 - The surface area of LI $\sim 1/30^{\text{th}}$ that of the SI.
- The colon is permanently colonized by extensive number and variety of bacteria ($\sim 10^{12}$ /g content)
 - capable of several metabolic reactions
- Although some drugs may be rendered inactive, bacterial metabolism of other drugs may give rise to more active or toxic products.

- Sulfasalazine, which is used in the treatment of ulcerative colitis, provides an interesting example of a drug whose metabolites represent the active pharmacological species.



- Recent interest has emerged to exploit enzymes produced by colonic bacteria to targeted drug delivery to this region
- Some biopharmaceutical importance:
 - Targeted drugs to colon